

Divide and Conquer - Searching & Basic Problems (With Code)

Reference: Topic 5.docx.pdf

Solutions implemented in Python

Question 1: Search an Element in a Sorted Array

```
def binary_search(arr, key):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid + 1
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1

n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter key: "))
pos = binary_search(arr, key)

if pos != -1:
    print(f"Element found at position {pos}")
else:
    print("Element not found")
-----
```

Question 2: Find the Index of a Given Number

```
def binary_search_index(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1

n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))
target = int(input("Enter target: "))
print(f"Index = {binary_search_index(arr, target)}")
-----
```

Question 3: Count the Number of Iterations in Binary Search

```
def binary_search_iterations(arr, key):
    low, high = 0, len(arr) - 1
    iterations = 0
    while low <= high:
        iterations += 1
        mid = (low + high) // 2
        if arr[mid] == key:
            print("Element found")
            print(f"Iterations = {iterations}")
            return
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    print("Element not found")
    print(f"Iterations = {iterations}")

n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter key: "))
binary_search_iterations(arr, key)
-----
```

Question 4: Find the First Occurrence of an Element

```
def first_occurrence(arr, key):
    low, high = 0, len(arr) - 1
    res = -1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            res = mid
            high = mid - 1 # Keep searching in the left half
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return res

n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter key: "))
idx = first_occurrence(arr, key)

if idx != -1:
    print(f"First occurrence at index {idx}")
else:
    print("Element not found")
-----
```

Question 5: Find the Last Occurrence of an Element

```
def last_occurrence(arr, key):
    low, high = 0, len(arr) - 1
    res = -1
```

```

while low <= high:
    mid = (low + high) // 2
    if arr[mid] == key:
        res = mid
        low = mid + 1 # Keep searching in the right half
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1
return res

n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter key: "))
idx = last_occurrence(arr, key)

if idx != -1:
    print(f"Last occurrence at index {idx}")
else:
    print("Element not found")
-----

```

Question 6: Highest and Lowest Temperature Recorded

```

def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        return (arr[low], arr[high]) if arr[low] < arr[high] else
(arr[high], arr[low])

    mid = (low + high) // 2
    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

n = int(input("Enter number of days: "))
arr = list(map(int, input("Enter temperatures: ").split()))
min_temp, max_temp = find_min_max(arr, 0, n - 1)

print(f"Minimum Temperature = {min_temp}")
print(f"Maximum Temperature = {max_temp}")
-----

```

Question 7: Best and Worst Student Marks

```

def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        return (arr[low], arr[high]) if arr[low] < arr[high] else
(arr[high], arr[low])

    mid = (low + high) // 2
    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

```

```

        return min(min1, min2), max(max1, max2)

n = int(input("Enter number of students: "))
arr = list(map(int, input("Enter marks: ").split()))
min_mark, max_mark = find_min_max(arr, 0, n - 1)

print(f"Minimum Mark = {min_mark}")
print(f"Maximum Mark = {max_mark}")
-----

```

Question 8: Stock Price Analysis

```

def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        return (arr[low], arr[high]) if arr[low] < arr[high] else
(arr[high], arr[low])

    mid = (low + high) // 2
    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

n = int(input("Enter number of days: "))
arr = list(map(int, input("Enter prices: ").split()))
min_price, max_price = find_min_max(arr, 0, n - 1)

print(f"Minimum Stock Price = {min_price}")
print(f"Maximum Stock Price = {max_price}")
-----

```

Question 9: Fastest and Slowest Race Times

```

def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        return (arr[low], arr[high]) if arr[low] < arr[high] else
(arr[high], arr[low])

    mid = (low + high) // 2
    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

n = int(input("Enter number of racers: "))
arr = list(map(int, input("Enter times: ").split()))
fastest, slowest = find_min_max(arr, 0, n - 1)

print(f"Fastest Time = {fastest}")
print(f"Slowest Time = {slowest}")
-----

```

Question 10: Largest and Smallest Building Heights

```
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        return (arr[low], arr[high]) if arr[low] < arr[high] else
(arr[high], arr[low])

    mid = (low + high) // 2
    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)

n = int(input("Enter number of buildings: "))
arr = list(map(int, input("Enter heights: ").split()))
min_h, max_h = find_min_max(arr, 0, n - 1)

print(f"Minimum Height = {min_h}")
print(f"Maximum Height = {max_h}")
-----
```